gpeyt1-25 · Geotextile length

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

Same 16 ft geotextile wall: $\gamma = 120$ pcf, $\varphi = 32°$, $q = 250$ psf, FS = 1.5. Design a layer at $Z = 4$ ft below the top with $S_v = 2.0$ ft. Interface friction $\delta = \frac{2}{3}\varphi$, $c_a = 0$. Wall length $L = L_e + L_R$ (L_R measured from the face to the Rankine plane).

Most nearly, the total geotextile length L at Z = 4 ft is:

A. 1.8 ft
B. 10.2 ft
C. 4.0 ft
D. 8.4 ft

Given equations / data

> $\sigma_h = K_a(\gamma Z + q)$; $L_e = S_v \sigma_h FS/[2(c_a + \gamma Z \tan\delta)]$; $L_R = (H - Z)\tan(45 - \varphi/2)$

Worked solution

1. $\sigma_h = 0.307(120\times4 + 250) = 224$ psf
2. $\delta = 21.33°$; $L_e = 2.0\times224\times1.5/[2\times120\times4\times\tan21.33] = 1.79$ ft
3. $L_R = (16 - 4)\tan(45 - 16) = 6.65$ ft
4. $L = 1.79 + 6.65 = 8.4$ ft (a practical minimum length would also be checked)

Final answer

8.4 ft

Why the other values are wrong

- **1.8 ft** — reported L_e only (forgot L_R)
- **10.2 ft** — used one face of friction (dropped the 2)
- **4.0 ft** — measured L_R with Z instead of $H - Z$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

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